Growth Rates

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A growth rate is a numerical measure of the rate of expansion or contraction of a quantity over time. Suppose Swift Sales Co. reported revenue of $100 million last year. This year, revenue grew by $12 million. The *growth rate* of revenue is defined as the ratio of the increase to the starting value, so in this case

$$g = \frac{12}{100} = .12 = 12\% \tag{1}$$

If the Swift Sales Co. were to grow at 12% per annum for 8 years, what would revenue be in the 8th year? This is steady growth. Starting with sales of $100 million in the first year, sales increase by 12% to $112 million in year two. For the third year, the starting revenue is now $112, and 12% of that is .12x112 = $13.44. The growth is *compounding* in that the growth in the third year applies both to the original base and the extra $12 million added in the second year.

More generally, the revenue at the end of N years of growth will be $100(1+g)^N = 100(1.12)^N$. So, the answer for Swift Sales is that revenue in year 8 will be

$$100G^8 = 100(1+g)^8 = 100(1.12)^8 = 247.6 \tag{2}$$

Compounded interest rates are growth rates. Given compounding—interest on prior interest—an interest rate is the growth rate of money invested or loaned at that interest rate. If a bank paid a 6% interest rate, your deposits would grow at 6% per annum (ignoring taxes). If a bank gave you a loan at a 10% interest rate, the balance outstanding would grow at 10% per annum if you didn't make any payments to reduce the balance.

There is an extra complication that arises in the arithmetic of interest rates. The custom in banking and finance is to state "nominal" rather than real interest rates. An example should make this clear. A bank quotes a mortgage loan at a nominal 8% rate. Mortgage interest is computed monthly. What the bank really means is that they will charge interest of (8/12)% each month—a monthly interest rate of 0.667%. See the trick? They don't take the twelfth root; instead they divide by 12. In this case, the *real* annual rate of interest is

$$(1+0.00667)^{12} - 1 = 0.08304\,.$$

So you are really paying 8.3% interest, not 8%. This is what credit card companies have to reveal when the fine print shows the APR interest (annual percentage rate).

To be fair, the same thing happens on the deposit side. If the bank pays nominal 5% interest, you have to look at the compounding. If it is compounded daily, even over weekends, then the real interest you are getting is

$$(1+0.05/365)^{365} - 1 = 0.0513 = 5.13\%\,.$$

What happens if you compound more and more frequently? Every day, or every second, or every nanosecond? Swiss mathematician Leonhard Euler worked out the answer in the 1700s.

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$$\lim_{N\to\infty}(1+g/N)^N = e^g \tag{3}$$

Where $e = 2.71828182845904$. Like π, Euler's ***e*** is a number that crops up over and over again. There are times when the math of continuous compounding—of continuous growth—is simpler than the math of year-to-year or period-to-period calculations.

In continuous time, a growing stream of cash flows would be represented as

$$c(t) = c_0 e^{gt} \tag{4}$$

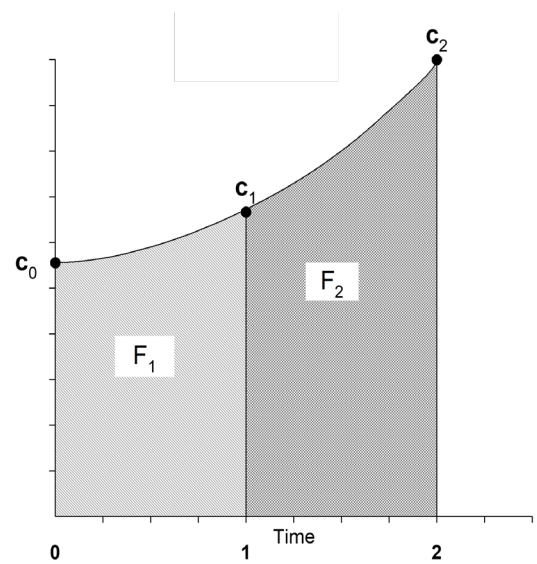
where g is *continuous* rate of growth and c_0 is the "instantaneous" rate of cash flow at time t=0. To understand the relationships between continuous and instantaneous rates study Figure 1.

In continuous time, a growing stream of cash flows is represented as

$$c(t) = c_0 e^{gt} \tag{5}$$

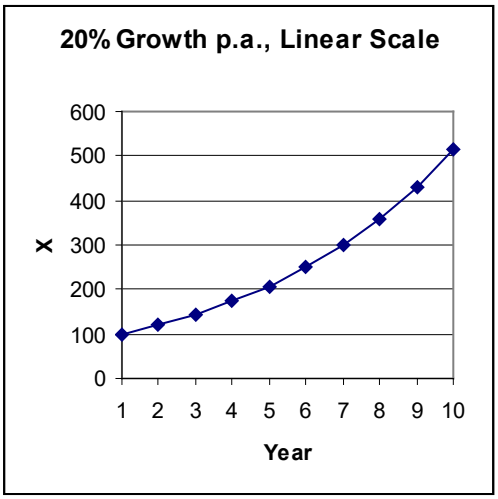
where g is the "continuous time" rate of growth and c_0 is the "instantaneous" rate of cash flow at time t=0. To understand the relationships between continuous and instantaneous rates study Exhibit 1.

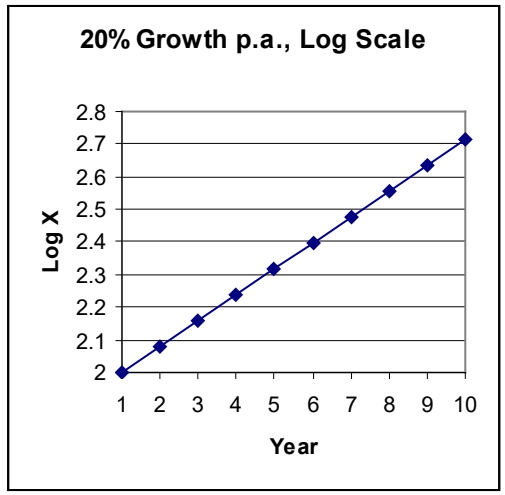
Exhibit 1—Continuous Growth



This diagram shows *c(t)*, an "instantaneous" rate of cash flow growing at rate *G*. The amount of cash flow observed over the first year (or period) is F_1 and the amount observed over the second year is F_2. Note that one never observes the instantaneous flows $c(t)$. Only its integrated value over a period is actually observed (e.g., we see revenues per year, not revenues per second). Given the annual flows F_1 and F_2, the continuous time quantities are:

Exhibit 2





$$G = \ln(F_2 / F_1) \tag{6}$$

$$c_0 = F_1 \frac{G}{e^G - 1} \tag{7}$$

$$c_1 = c_0 e^G \tag{8}$$

We convert from a continuous flow model to growth in annual flows by using this conversion rule

Here is the conversion rule:

If

$$c_0 e^G = c_0 (1+g)$$

then it must be that

$$G = \ln(1+g)\,. \tag{9}$$

Returning to the problem of Swift's growth in sales, its 12% annual growth, compounded annually, is "equivalent" to fully continuous growth at the rate $G = \ln(1.12) = 0.113329$. The continuous growth rate is less because it compounds continually. Still, the numbers work out the same for t = 1, 2, 3 and so on.

Let's generalize. If we start with a quantity X_0 at time 0, then after t years of steady growth at rate g per year, the quantity will be

$$X(t) = X_0 G^t = X_0 (1+g)^t \tag{10}$$

This sort of steady growth is called exponential because the value of X is rising exponentially with t (technically, t is an exponent).

It is often useful to plot a graph of a growing quantity on a logarithmic scale. This is because the growing quantity can increase so much that a normal linear graph only shows the most recent data

points; the rest are compressed down near the x-axis. Take the natural logarithm of both sides of equation (10):

$$\log X = \ln X_0 + tG = \ln(X_0) + t\ln(1+g). \tag{11}$$

This equation is linear in t, not exponential. A plot of $\ln(X)$ vs. t will be a straight line. Importantly, note that the slope of the line is G, not g.

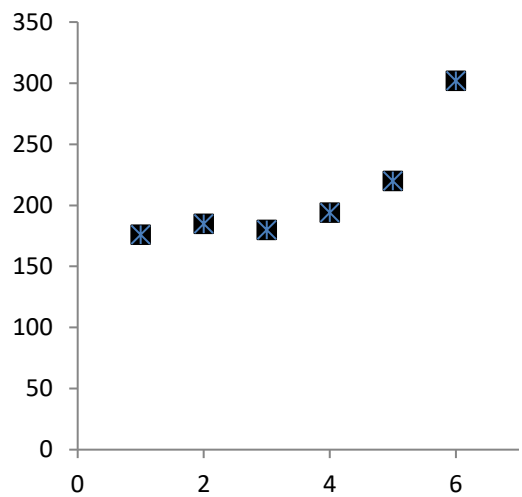
Estimating Growth Rates

Thus far we have discussed steady growth. What if growth is not steady, but jumps around from period to period? Most real-world data exhibits non-steady growth. Nevertheless, we are often asked to estimate the growth rate implicit in such data.

Suppose we are given six years of sales data shown in and wish to determine the "average" rate of growth underlying that data. Six years of sales data gives five observations of annual growth rates, as shown in Exhibit 3.

Exhibit 3—Six Years of Sales

t	X(t)	g(t)
1	176	
2	185	5.1%
3	180	-2.7%
4	194	7.8%
5	220	13.4%
6	302	37.3%



The first, seemingly obvious, way to proceed is to average the five observed growth rates:

$$g_{\text{avg}} = 5.1 + (-2.7) + 7.8 + 13.4 + 37.3)/5 = 12.2\%.$$

The second standard answer is to compute the geometric average. Technically, the geometric average is

$$g_{\text{geo}} = [(1.051)(.973)(1.087)(1.134)(1.373)]^{1/5} - 1 = 11.4\%, \tag{12}$$

but, in practice, it is much simpler to compute g_{geo} by just working with the two endpoints, computing

$$g_{\text{geo}} = \left(\frac{302}{176}\right)^{1/5} - 1 = 11.4\%. \tag{13}$$

These two ways of computing an average growth rate produce different results and people will ask "which one is right?" To make this judgment, one must stop looking at these as growth rates and adopt the language of statistics. Both numbers are *estimates* of hidden parameters in a model of growth. These estimates are, in turn, the *answers* to hidden questions. The first pair of question-answers is this: Suppose that the quantity being studied is labeled X_t. Suppose further that the each year's growth is a hidden constant m plus an additive random error ε_t. That is,

$$\frac{X_t}{X_{t-1}} = (1 + g_t) = (1 + m + \varepsilon_t) \tag{14}$$

If the errors themselves have a mean of zero and are independently distributed (uncorrelated from one observation to another), then the best estimate of *m* is g_{avg}. This is the case because the errors add to the hidden mean and taking the average, averages out the errors. So, g_{avg} is the right answer to that question.

On the other hand, suppose that each year's growth rate is a constant times a multiplicative error. In, particular,

Suppose, instead, you believe each year's growth rate is a multiplicative deviation

$$\frac{X_{t+1}}{X_t} = (1 + g_t) = (1 + m)E_t \tag{15}$$

In this case, it is helpful to define $\mu = \ln(1 + m)$ and re-write (15) like this:

$$\frac{X_t}{X_{t-1}} = e^{\mu} e^{\varepsilon_t} = e^{\mu + \varepsilon_t}. \tag{16}$$

If the now-additive errors have a mean of zero and are independent, the best estimate of e^{μ} is the geometric mean g_{geo}. So, again we have a commonly used number which is the answer to a hidden question about a hidden parameter.

In the case of the geometric mean, there is a subtle complexity—estimating e^{μ} is ***not*** the same thing as estimating the expected value of $1 + g_t$. The issue is that the error term ε_t is in the exponent. Even if it has a mean of zero, positive errors will be magnified by the exponentiation and negative errors will be squelched. Thus, if the variance of errors is σ_ε^2, then the expected value of e^{ε_t} turns out to be $e^{\sigma_\varepsilon^2/2}$. This extra nugget—a factor due to the variance in growth rates—is the reason that the arithmetic and geometric averages are different. That is,

$$g_{\text{avg}} = g_{\text{geo}} e^{\sigma^2/2} \approx g_{\text{geo}} + \sigma^2/2 \tag{17}$$

If σ_ε^2 were zero, they would be the same. In the case at hand, g_{avg} is 0.8% higher than g_{geo}. An estimate of $\sigma_\varepsilon^2/2$ is 0.83% (half of the variance of $\ln[g_t]$): it is very close to the gap between the arithmetic and geometric averages.

The uncomfortable truth is that although the geometric average is the better estimate of past growth, it may underestimate expected future growth. The expected future growth should include the non-linear effects of the variability in growth. The arithmetic average does just that.

But, there is more. If the errors are not completely uncorrelated, then the arithmetic average may seriously overestimate future growth. For example, over the 75 years from 1926 to 2001, the arithmetic average return of the S&P 500 index has been 12.5% whereas the geometric average has been 10.5%. The variance in returns has been about 4%, which is just twice the difference. If the future is like the past, and if errors are uncorrelated, then we might expect future returns to compound at 12.5%. But, if you believe, as I do, that stock-price returns are mildly negatively correlated over four-to-seven year periods, then eras of high return tend to be followed by eras of lower returns. That means that the arithmetic average is a serious overestimate.

And, there is more. These two numbers, g_{avg} and g_{geo}, are not the truth. They are estimates. The errors in these estimates can itself be estimated and those errors compound over long times. Thus, over long-horizons, and even if errors are independent, g_{avg} overestimates future growth and g_{geo} underestimates it.[1]

If we are given a long data series, the arithmetic and geometric averages are both useful ways to estimate growth rates. However, given a short series, as in Exhibit 3's six years of data on sales, there is a valid concern about outliers. Here, in this particular case, the last year's growth rate was especially high. This kicks up the geometric average because it is simply based on the ratio of the last year's sales to the first year's sales. One way to diminish the impact of the last observation on a growth estimate is to consider a third model.

$$X_t = X_0(1+m)^t e^{\varepsilon_t} \tag{18}$$

This model postulates a non-random steady growth at rate m, combined with errors that effect each observation of *X*. That is, the noise is in our measures of sales. Taking logs, (18) becomes

$$\ln[X_t] = a + t(1+m) + \varepsilon_t \tag{19}$$

where $a = \ln[X_0]$. Thus, estimating $1+m$ is a linear regression problem. Excel's convenient function LOGEST does this in one step:

$$\text{LOGEST}\left(\{176,185,180,194,220,302\}\right) = 1.0987$$

The full regression fit is shown in Exhibit 4 (Excel function GROWTH). This method of estimating a growth rate is not appropriate for the prices of assets, but does make sense for relatively short duration data based on accounting number.

A fourth method of estimating the growth rate of this sales data is the easiest—take the median. For an odd number of data points, this is the middle ranked number. For an even number of data points, it is half-way between the two middle ranked numbers. Excel's convenient function yields

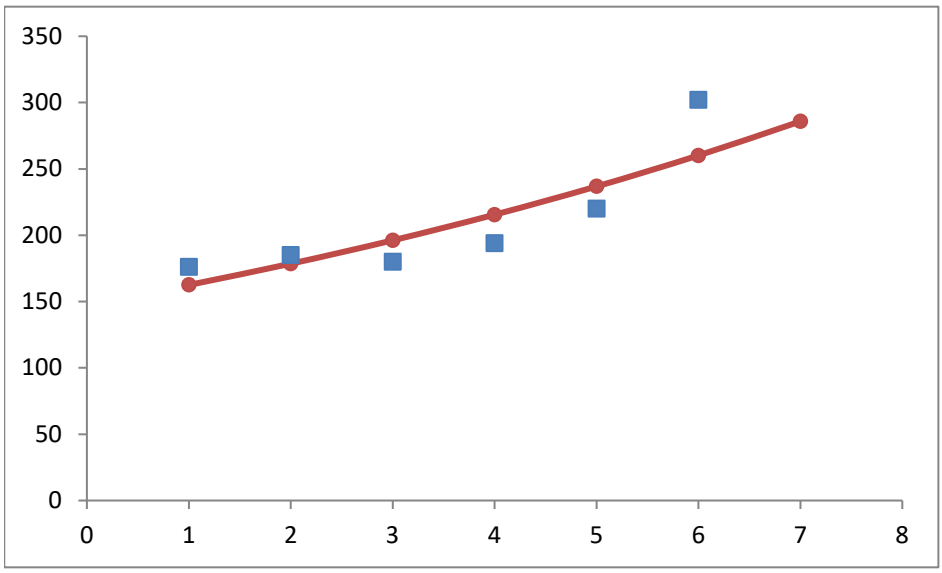
$$\text{MEDIAN}\left(\{5.1,-2.7,7.78,13.4,37.27\}\right) - 1 = 7.78\%,$$

[1]Jacquier, Eric, Alex Kane, and Alan J. Marcus, "Geometric or Arithmetic Mean: A Reconsideration." *Financial Analysts Journal*, Vol. 59, 2003, pp. 46-53.

by far the lowest estimate of the four we have considered. The virtue of the median is that it is unswayed by large positive and negative values. If you believe these are errors in measurement, rather than fantastically good or bad performance, the median is the right measure.

Interestingly, the median is also the best estimate if you believe that the underlying errors are distributed according to the little-known Laplace or symmetric-exponential distribution. You might be inclined to make this assumption because this is pretty-much how the growth rates of large

Exhibit 4—X(t) and Regression Fit



corporations are distributed. Despite years of assuming that the $\ln(g_t)$ is normally distributed, recent empirical work shows that it is not. The distribution of the logs of real corporate growth rates is both more concentrated around the median and, at the same time, has fatter tails than the normal.[2]

[2] Bottazzi, Giuylio,and Angelo Secchi, "Explaining the Distribution of Firm Growth Rates," *The RAND Journal of Economics*, Vol. 37, 2006, pp. 235-256.